

a. Introduction to WAN Technologies

Definition of WAN

A Wide Area Network (WAN) is a telecommunications network that spans a large geographic area — often a city, country, or multiple countries — connecting Local Area Networks (LANs) that are otherwise physically separate. Unlike a LAN, a WAN typically relies on links leased or purchased from a third-party telecom service provider, since an organization does not own the physical infrastructure covering such large distances. WANs are what allow branch offices, data centers, and remote users to communicate with one another as though they were on the same local network.

Task 1: Create a table comparing at least three types of WAN connections with columns for speed, cost, and typical use cases.

WAN Connection Type	Typical Speed	Relative Cost	Typical Use Case
Leased Line	Symmetric, dedicated — from 64 kbps up to several Gbps (fibre)	High (fixed monthly, pay for reserved bandwidth regardless of use)	Mission-critical enterprise backbone links needing guaranteed, always-on bandwidth (e.g., HQ-to-datacenter link)
DSL (ADSL/VDSL)	DSL gives you uneven (asymmetric), non-dedicated (shared) bandwidth, with ADSL topping out around 24 Mbps and VDSL reaching up toward 100 Mbps under good conditions	Low	Small office/home office internet access, branch offices with modest bandwidth needs
MPLS (Multiprotocol Label Switching) is a WAN technology that speeds up and controls how traffic is routed across a provider's network, using labels instead of full IP-address lookups at every hop.	Scalable - MPLS bandwidth isn't fixed to one tier the way a basic DSL line is. MPLS is actively managed by the telecom provider. — typically 2 Mbps to several Gbps per site	Moderate-High (priced per site/class of service, not flat like a leased line)	Multi-site enterprise WANs needing QoS for VoIP, video conferencing, and ERP traffic
Metro Ethernet	10 Mbps to 10 Gbps, Ethernet-based	Moderate	Connecting offices/data centers within the same metro area needing high, scalable bandwidth

Task 2: Draw a labeled diagram showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2-3 sentences.

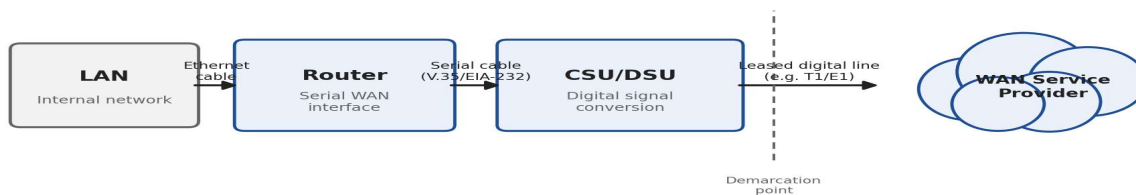


Fig. 1 — Router-to-WAN connection via a CSU/DSU to a leased digital line

The router generates digital data but cannot send it directly onto a telecom carrier's line, so it hands that data to the CSU/DSU over a serial WAN interface.

The CSU (Channel Service Unit) terminates the digital circuit, protects it from electrical interference, and handles line conditioning and diagnostics, while the DSU (Data Service Unit) converts the router's data into the framed, clocked signal format the carrier's line requires (and back again for incoming data). The WAN service provider's network then carries this signal — over the leased line, e.g., a T1/E1 circuit — to the destination site, where an identical CSU/DSU and router perform the reverse conversion.

In short:

- **CSU** = manages and protects the *physical connection to the carrier's line* (termination, protection, diagnostics)
- **DSU** = manages the *data itself* (conversion, framing, timing between router and line)

Modern routers — have the CSU/DSU functionality **integrated directly into a WAN interface card (WIC or HWIC)** that plugs into the router chassis.

Task 3: List three real-world examples of popular apps or services that rely on WAN connections, and briefly describe how WAN enables their functionality.

App / Service	How WAN Enables It
IRCTC	IRCTC's booking servers in its data centers must be reachable from millions of user devices across India. WAN links (leased lines/MPLS connecting IRCTC's data centers to ISPs, plus the public internet backbone) carry booking requests, seat-availability queries, and payment traffic between distributed users and the centralized reservation system in real time.
WhatsApp	WhatsApp's servers are hosted in data centers that may be far from any given user. Every message, call, or media file travels from the sender's device, across the ISP's access network and the wider internet (a global WAN), to WhatsApp's servers and back down to the recipient — WAN connectivity is what makes this cross-city or cross-country delivery possible within seconds.
Zomato	Zomato's restaurant listings, live order tracking, and delivery-partner location updates all depend on continuous communication between customer apps, restaurant-partner apps, delivery-partner apps, and Zomato's backend servers. This many-to-one, geographically spread communication is carried over WAN links connecting each city's users to Zomato's centralized (or regionally distributed) cloud infrastructure.

Task 4: Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device.

Device: Cisco ASR 1000 Series Aggregation Services Router — an enterprise-grade WAN aggregation router deployed at the WAN/internet edge to consolidate and secure traffic from multiple branch or access links.

Official datasheet: <https://www.cisco.com/c/en/us/products/collateral/routers/asr-1000-series-aggregation-services-routers/datasheet-c78-731632.html>



Two key features:

- Purpose-built QuantumFlow Processor architecture: the ASR 1000 Series was the first Cisco platform to use the Cisco QuantumFlow Processor, a chip built specifically for edge-based service delivery, letting it accelerate advanced features such as NAT, Zone-Based Firewall, and Deep Packet Inspection while aggregating WAN links.
- Modular, scalable hardware: depending on the model (ASR 1002 through ASR 1006), the router offers built-in Gigabit and 10-Gigabit Ethernet ports, software redundancy, and slots for Shared Port Adapters/Ethernet Port Adapters — allowing the same platform family to scale from a compact branch-aggregation box up to a large chassis-based data-center or service-provider aggregation router.